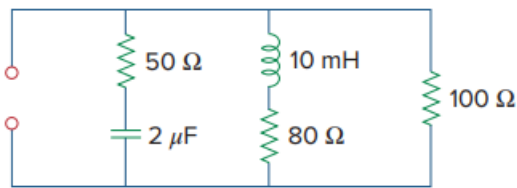


Problem

The network in Fig. 9.87 is part of the schematic describing an industrial electronic sensing device. What is the total impedance of the circuit at 4 kHz?

Figure 9.87:



Step-by-step solution

Step 1 of 4

Refer to Figure 9.87 in the text book.

Calculate the value of the reactance due to inductance.

$$X_L = j\omega L$$

Substitute $2\pi f$ for ω .

$$X_L = j2\pi fL$$

Substitute 4 kHz for f and 10 mH for L .

$$\begin{aligned} X_L &= j2\pi(4\text{ k})(10\text{ m}) \\ &= j251.33\ \Omega \end{aligned}$$

Therefore, the value of the reactance X_L is, $j251.33\ \Omega$.

Step 2 of 4

Calculate the value of the reactance due to capacitance.

$$X_C = \frac{1}{j\omega C}$$

Substitute $2\pi f$ for ω .

$$X_C = \frac{1}{j2\pi fC}$$

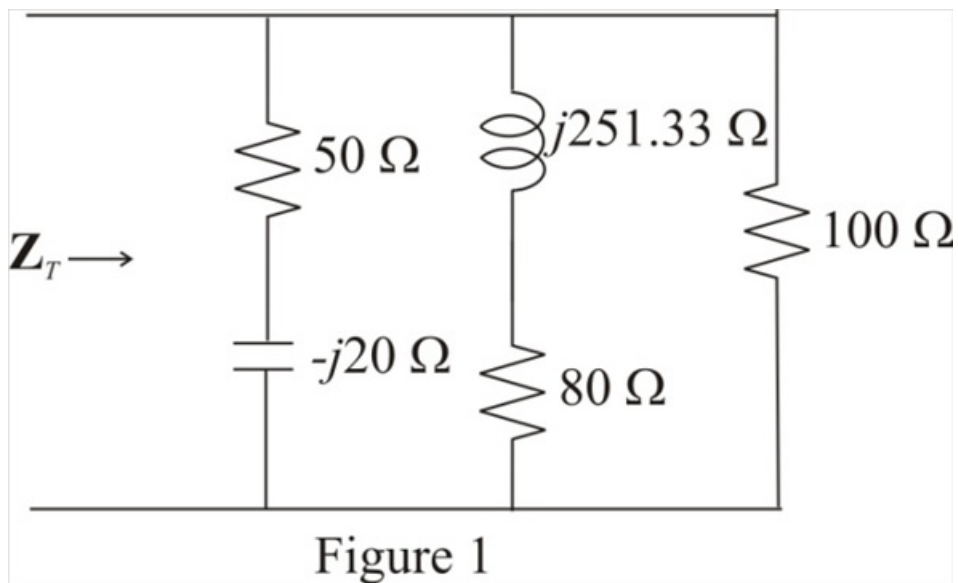
Substitute 4 kHz for f and $2\ \mu\text{F}$ for C .

$$\begin{aligned} X_C &= \frac{1}{j2\pi(4\text{ k})(2\ \mu)} \\ &= \frac{1}{j0.05} \\ &= -j20\ \Omega \end{aligned}$$

Therefore, the value of the reactance X_C is, $-j20\ \Omega$.

Step 3 of 4

Consider the circuit diagram shown in Figure 1.



Step 4 of 4

Calculate the value of the impedance $\mathbf{Z}_T$.

$$\begin{aligned}\mathbf{Z}_T &= (50 - j20) \parallel (80 + j251.33) \parallel 100 \\ &= \frac{(50 - j20)(80 + j251.33)(100)}{(50 - j20)(80 + j251.33) + (100)(50 - j20) + (80 + j251.33)(100)} \\ &= \frac{902660 + j1096650}{9026.6 + j10966.5 + 5000 - j2000 + 8000 - j25133} \\ &= \frac{902660 + j1096650}{22026.6 + j34099.5} \\ \mathbf{Z}_T &= 34.76 - j4.02 \\ &= 34.98 \angle -6.6^\circ \Omega\end{aligned}$$

Therefore, the value of the total impedance $\mathbf{Z}_T$ is, $34.98 \angle -6.6^\circ \Omega$.